Shriya Arunkumar

Autonomous Navigation | Robotic Perception | Reconfigurable Robotics

arunkumarshriya@gmail.com | Bengaluru, India
github.com/Shriya-Arunkumar | LinkedIn: shriya-arunkumar

EDUCATION

PES University, Bengaluru | Expected graduation: 2028
B.Tech, Electronics and Communication Engineering | CGPA: 8.50/10
Minor in Computer Science | CGPA: 9.00/10

RESEARCH EXPERIENCE

Summer Research Intern - cRAIS, PES University | Summer 2026
Human-Robot Interaction, Gaze Estimation, Robot Attention

Developed a webcam-based multi-person robot attention software prototype connecting face and gaze estimation, object-level attention reasoning, confidence-aware inference, and attention arbitration to ROS2 and Gazebo robot-eye simulation. Further algorithm refinement and experimental evaluation are planned.

Research Intern - cRAIS, PES University | Current
Working on a tri-modal robotic platform exploring mobility across land, air, and water. Focus: robotic systems design, CAD development, mechanical integration, and experimental research.

SELECTED TECHNICAL PROJECTS

Multi-Person Robot Attention System | Prototype implemented
Integrated visual perception and attention selection with ROS2 Humble and Gazebo Classic. Python, OpenCV, MediaPipe, and C++ support the perception-to-simulation pipeline. Quantitative evaluation remains planned.

Trimodal Robot - Ongoing Research | Design and development in progress
An ongoing robotics research project investigating a robotic platform designed to operate across terrestrial, aerial, and aquatic environments. Further technical details are not publicly disclosed at this stage.

Autonomous Navigation for Indian Roads | Research in progress
Primary research responsibility: investigating adaptive multi-sensor fusion under varying sensor reliability and environmental uncertainty. Algorithm development is ongoing; simulation and experimental evaluation are planned.

Custom 5-DOF Robotic Arm | Under development
Designing an SO-101-inspired 3D-printed arm with an actuated gripper, modular mechanisms, and an Arduino/PCA9685 control architecture. Joint feedback, ROS2, and robot learning are future directions.

Dynamic Vision Sensor Simulator | Software demonstration
Python exploration of per-pixel brightness changes as event activity. Public source and demonstration are linked from the portfolio; physical-sensor benchmarking is not reported.

TECHNICAL SKILLS

Programming: Python, C++ | **Vision:** OpenCV, MediaPipe, gaze estimation, facial landmarks
Robotics: ROS2 Humble, Gazebo Classic, simulation integration
Design: Fusion 360, SolidWorks, mechanical assemblies, 3D modelling
Embedded: Arduino, servo control, electronics prototyping
Research: Literature review, technical documentation; developing evaluation and analysis methods

SELECTED ACHIEVEMENTS

DVS simulator: public software and demonstration. Research project documentation available at Shriya-Arunkumar.github.io. No peer-reviewed publications listed.

Parallax 3.0: 1st, 2025 | Hacksimulink: 2nd, September 2026
Math Hackathon: 5th, 2025 | Riverie Hacks: 7th, 2024
NOVA Residency: pitched a deeptech startup idea to 20+ investors.
PES Women's Basketball Team: 3rd, Revels Cup 2026, MIT Manipal.